



GNU nano 8.7.1

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

int main() {
    char command[100];

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid_t pid = fork();

    if (pid < 0) {
        printf("Fork failed!\n");
        return 1;
    }

    else if (pid == 0) {
        printf("Child Process\n");
        printf("Child PID : %d\n", getpid());

        printf("Executing command: %s\n", command);

        execlp(command, command, NULL);

        printf("Command execution failed!\n");
    }

    else {
        printf("Parent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        wait(NULL);

        printf("Child process completed.\n");
    }

    return 0;
}
```

```
navya@DESKTOP-LST06QP:/mnt/c/Windows/system32$ mkdir OS_LAB
navya@DESKTOP-LST06QP:/mnt/c/Windows/system32$ cd OS_LAB
navya@DESKTOP-LST06QP:/mnt/c/Windows/system32$ nano task1.c
navya@DESKTOP-LST06QP:/mnt/c/Windows/system32$ gcc task1.c -o task1
navya@DESKTOP-LST06QP:/mnt/c/Windows/system32$ sudo apt update
```

```
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [32.8 kB]
Hit:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:6 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.1 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [62.2 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [161 kB]
Fetched 574 kB in 4s (143 kB/s)
25 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

```
navya@DESKTOP-LST06QP:/mnt/c/Windows/system32$ sudo apt install build-essential
```

```
build-essential is already the newest version (12.12ubuntu2.26.04.2).
```

```
Summary:
```

```
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 25
```

```
navya@DESKTOP-LST06QP:/mnt/c/Windows/system32$ ./task1
```

```
Enter a Linux command: ls
```

```
Parent Process
```

```
Parent PID : 1688
```

```
Child Process
```

```
Child PID : 1689
```

```
Child PID : 1689
```

```
Executing command: ls
```

```
task1 task1.c
```

```
Child process completed.
```

```
navya@DESKTOP-LST06QP: /mnt/c/Windows/system32$ uname -a
Linux DESKTOP-LST06QP 5.15.0-56-generic #62-Ubuntu SMP Tue Aug 14 22:03:11 UTC 2023 x86_64 x86_64 x86_64 GNU/Linux
navya@DESKTOP-LST06QP: /mnt/c/Windows/system32$ lscpu
Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
Address sizes: 48 bits physical, 48 bits virtual
Byte Order: Little Endian
CPU(s): 8
On-line CPU(s) list: 0-7
Vendor ID: AuthenticAMD
Model name: AMD Ryzen 5 7235HS
CPU family: 25
Model: 68
Thread(s) per core: 2
Core(s) per socket: 4
Socket(s): 1
Stepping: 1
BogoMIPS: 6387.81
Flags: fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx mmxext fxsr_opt
pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid extd_apicid tsc_known_freq pni pclmulqdq
ssse3 fma cx16 sse4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm cmp_legacy svm cr8_legacy abm sse4a mis
alignsse 3dnowprefetch osvw topoext perfctr_core ssbd ibrs ibpb stibp vmmcall fsgsbase bmi1 avx2 smep bmi2 erms invpcid rdseed
adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves clzero xsaveerptr arat npt nrip_save tsc_scale vmcb_clean flushb
yasid decodeassists pausefilter pfthreshold v_vmsave_vmload umip vaes vpclmulqdq rdpid fsrm
Virtualization features:
Virtualization: AMD-V
Hypervisor vendor: Microsoft
Virtualization type: full
Caches (sum of all):
L1d: 128 KiB (4 instances)
L1i: 128 KiB (4 instances)
```

```
L2:                2 MiB (4 instances)
L3:                8 MiB (1 instance)
NUMA:
  NUMA node(s):    1
  NUMA node0 CPU(s): 0-7
Vulnerabilities:
  Gather data sampling: Not affected
  Ghostwrite:         Not affected
  Indirect target selection: Not affected
  Itlb multihit:      Not affected
  L1tf:               Not affected
  Mds:                Not affected
  Meltdown:          Not affected
  Mmio stale data:    Not affected
  Old microcode:      Not affected
  Reg file data sampling: Not affected
  Retbleed:           Not affected
  Spec rstack overflow: Vulnerable: Safe RET, no microcode
  Spec store bypass:  Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:         Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:         Mitigation; Retpolines; IBPB conditional; IBRS_FW; STIBP always-on; RSB filling; PBRSE-eIBRS Not affected; BHI Not affected
  Srbds:              Not affected
  Tsx:                Vulnerable: No microcode
  Tsx async abort:    Not affected
  Vmscape:            Not affected
```

```
navy@DESKTOP-LST06QP:/mnt/c/Windows/system32$ ps
```

```
1418 pts/0    00:00:00 bash
1750 pts/0    00:00:00 ps
```

```
navy@DESKTOP-LST06QP:/mnt/c/Windows/system32$ ps -ef
```

```
root      1      0  0 14:51 ?        00:00:00 /sbin/init
root      2      1  0 14:51 hvc0    00:00:00 /init
root      7      2  0 14:51 hvc0    00:00:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root     47      1  0 14:51 ?        00:00:00 /usr/lib/systemd/systemd-journald
systemd+ 79      1  0 14:51 ?        00:00:00 /usr/lib/systemd/systemd-resolved
root     83      1  0 14:51 ?        00:00:00 /usr/lib/systemd/systemd-udevd
root    153      1  0 14:51 ?        00:00:00 snapfuse /var/lib/snapd/snaps/core22_2411.snap /snap/core22/2411 -o ro,nodev,allow_other,suid
root    154      1  0 14:51 ?        00:00:02 snapfuse /var/lib/snapd/snaps/snapd_27591.snap /snap/snapd/27591 -o ro,nodev,allow_other,suid
root    158      1  0 14:51 ?        00:00:00 snapfuse /var/lib/snapd/snaps/tree_54.snap /snap/tree/54 -o ro,nodev,allow_other,suid
```

```
root      184      1 0 14:51 ?      00:00:00 /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -n -F 1
root      185      1 0 14:51 ?      00:00:00 /usr/sbin/cron -f -P
message+  186      1 0 14:51 ?      00:00:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
root      190      1 0 14:51 ?      00:00:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
root      193      1 0 14:51 ?      00:00:00 /snap/snapd/current/usr/lib/snapd/snapd
root      194      1 0 14:51 ?      00:00:00 /usr/lib/systemd/systemd-logind
syslog    212      1 0 14:51 ?      00:00:00 /usr/sbin/rsyslogd -n -iNONE
root      229      1 0 14:51 tty1    00:00:00 /usr/sbin/agetty --noproset --noclear --issue-file=/etc/issue:/etc/issue.d:/run/issue.d:/usr/lib/issue.d
root      245      1 0 14:51 ?      00:00:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
_chrony   250     184 0 14:51 ?      00:00:00 /usr/sbin/chronyd -n -F 1 -x
_chrony   263     250 0 14:51 ?      00:00:00 /usr/sbin/chronyd -n -F 1 -x
root      388      2 0 14:51 ?      00:00:00 login -- navya
navya     429      1 0 14:51 ?      00:00:00 /usr/lib/systemd/systemd --user
navya     431     429 0 14:51 ?      00:00:00 (sd-pam)
navya     459     388 0 14:51 pts/1    00:00:00 -bash
root     1305      1 0 15:22 ?      00:00:00 /usr/libexec/wsl-pro-service
root     1415      2 0 15:24 ?      00:00:00 /init
root     1416    1415 0 15:24 ?      00:00:00 /init
navya     1418    1416 0 15:24 pts/0    00:00:00 -bash
polkitd   1556      1 0 15:25 ?      00:00:00 /usr/lib/polkit-1/polkitd --no-debug --log-level=notice
navya     1760    1418 0 15:31 pts/0    00:00:00 ps -ef
```

navya@DESKTOP-LST060P:/mnt/c/Windows/system32\$ top

```
Tasks: 30 total, 1 running, 29 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 5769.6 total, 5028.0 free, 539.4 used, 345.6 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 5230.2 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	24560	15604	11544	S	0.0	0.3	0:00.80	systemd
2	root	20	0	3180	2212	2072	S	0.0	0.0	0:00.01	init-systemd(Ub
7	root	20	0	3180	2044	1940	S	0.0	0.0	0:00.00	init
47	root	19	-1	50416	16944	15640	S	0.0	0.3	0:00.24	systemd-journal
79	systemd+	20	0	22416	14440	11944	S	0.0	0.2	0:00.09	systemd-resolve
83	root	20	0	35600	12568	9252	S	0.0	0.2	0:00.77	systemd-udev
153	root	20	0	159348	1864	1540	S	0.0	0.0	0:00.00	snappyfuse

```
top - 15:33:18 up 41 min, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 30 total, 1 running, 29 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 5769.6 total, 5009.5 free, 557.9 used, 345.7 buff/cache
```

MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 5211.4 avail Mem

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	24560	15604	11544	S	0.0	0.3	0:00.80	systemd
2	root	20	0	3180	2212	2072	S	0.0	0.0	0:00.01	init-systemd(Ub
7	root	20	0	3180	2044	1940	S	0.0	0.0	0:00.00	init
47	root	19	-1	50416	16944	15640	S	0.0	0.3	0:00.24	systemd-journal
79	systemd+	20	0	22416	14440	11944	S	0.0	0.2	0:00.09	systemd-resolve
83	root	20	0	35600	12568	9252	S	0.0	0.2	0:00.77	systemd-udevd
153	root	20	0	159348	1864	1540	S	0.0	0.0	0:00.00	snappfuse
154	root	20	0	454144	12376	1516	S	0.0	0.2	0:02.10	snappfuse
158	root	20	0	159348	1880	1556	S	0.0	0.0	0:00.00	snappfuse
184	root	20	0	2888	1952	1820	S	0.0	0.0	0:00.05	chronyd-starter
185	root	20	0	4472	3108	2860	S	0.0	0.1	0:00.01	cron
186	message+	20	0	8824	5376	4676	S	0.0	0.1	0:00.06	dbus-daemon
190	root	20	0	44096	29704	14556	S	0.0	0.5	0:00.25	networkd-dispat
193	root	20	0	2220748	38156	24808	S	0.0	0.6	0:00.82	snappd
194	root	20	0	18436	9472	8228	S	0.0	0.2	0:00.09	systemd-logind
212	syslog	20	0	220548	5996	4632	S	0.0	0.1	0:00.08	rsyslogd
229	root	20	0	5492	2748	2564	S	0.0	0.0	0:00.00	agetty
245	root	20	0	123460	32568	17956	S	0.0	0.6	0:00.22	unattended-upgr
250	_chrony	20	0	23732	11096	7100	S	0.0	0.2	0:00.08	chronyd
263	_chrony	20	0	12096	2432	1792	S	0.0	0.0	0:00.00	chronyd
388	root	20	0	8644	5556	4716	S	0.0	0.1	0:00.01	login
429	navya	20	0	22772	13464	10988	S	0.0	0.2	0:00.14	systemd
431	navya	20	0	22916	4096	2096	S	0.0	0.1	0:00.00	(sd-pam)
459	navya	20	0	6136	5412	3864	S	0.0	0.1	0:00.02	bash
1305	root	20	0	1857908	15260	12552	S	0.0	0.3	0:00.07	wsl-pro-service
1415	root	20	0	3188	1124	980	S	0.0	0.0	0:00.00	SessionLeader
1416	root	20	0	3204	1260	1108	S	0.0	0.0	0:00.03	Relay(1418)
1418	navya	20	0	6268	5540	3872	S	0.0	0.1	0:00.07	bash
1556	polkitd	20	0	306444	8440	7628	S	0.0	0.1	0:00.08	polkitd
1770	navya	20	0	8172	6008	3900	R	0.0	0.1	0:00.05	top